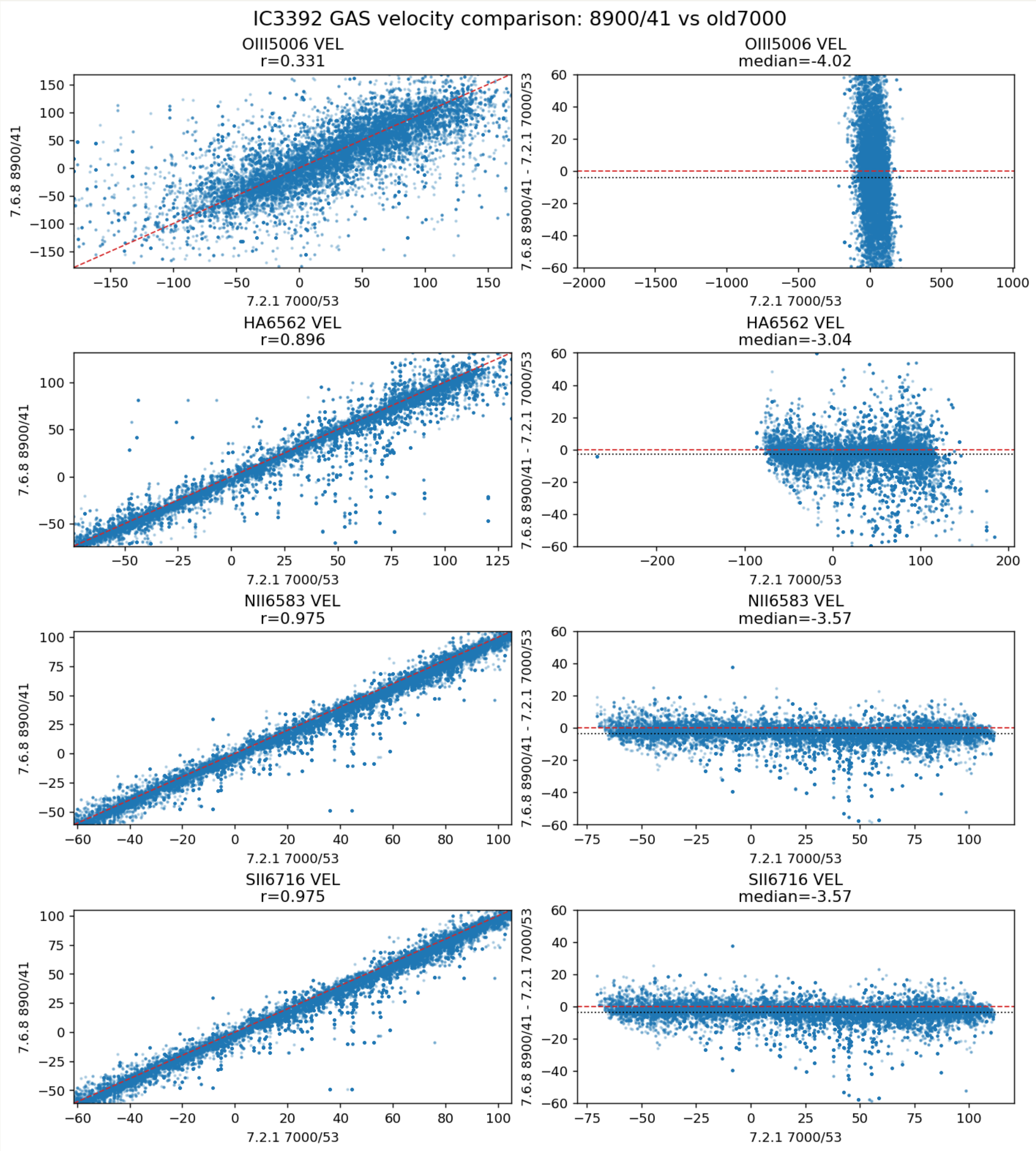
20260506 IC3392 GAS kinematics: VELSCALE=40 and VELSCALE=20 tests

Here we compares IC3392 GAS kinematics across the old 7000-A reference and the full 9100-A v7.6.8 tests with different log-rebin velocity scales.

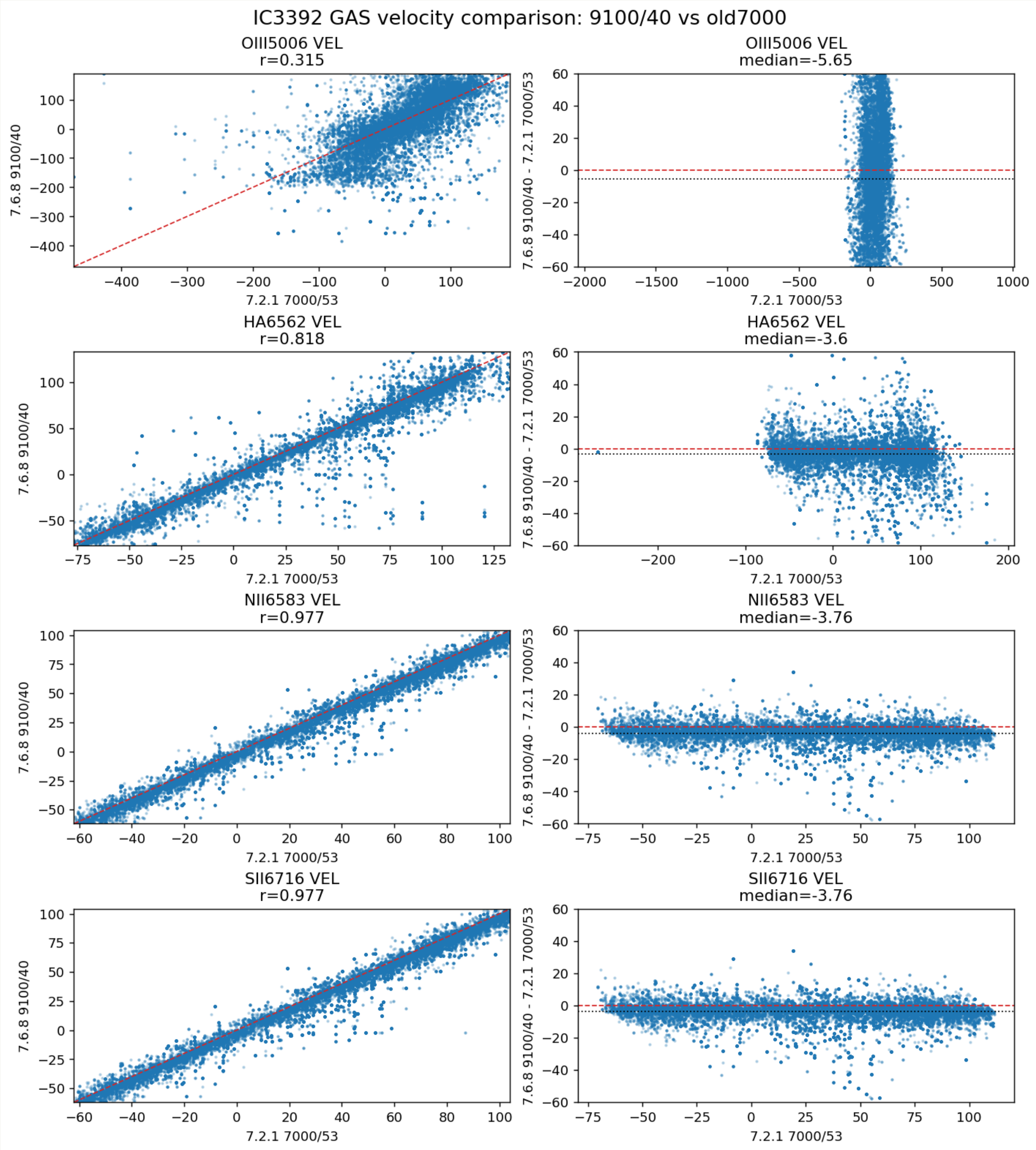
Runs compared here:

- `old_v721_7000`: original v7.2.1 7000-A reference, `VELSCALE = 53`
- `v768_8900_41`: v7.6.8 full-red-line setup to 8900 A, `VELSCALE = 41`
- `v768_9100_41`: v7.6.8 full-red-line setup to 9100 A, `VELSCALE = 41`
- `v768_9100_40`: v7.6.8 full-red-line setup to 9100 A, `VELSCALE = 40`
- `v768_9100_20`: v7.6.8 full-red-line setup to 9100 A, `VELSCALE = 20`

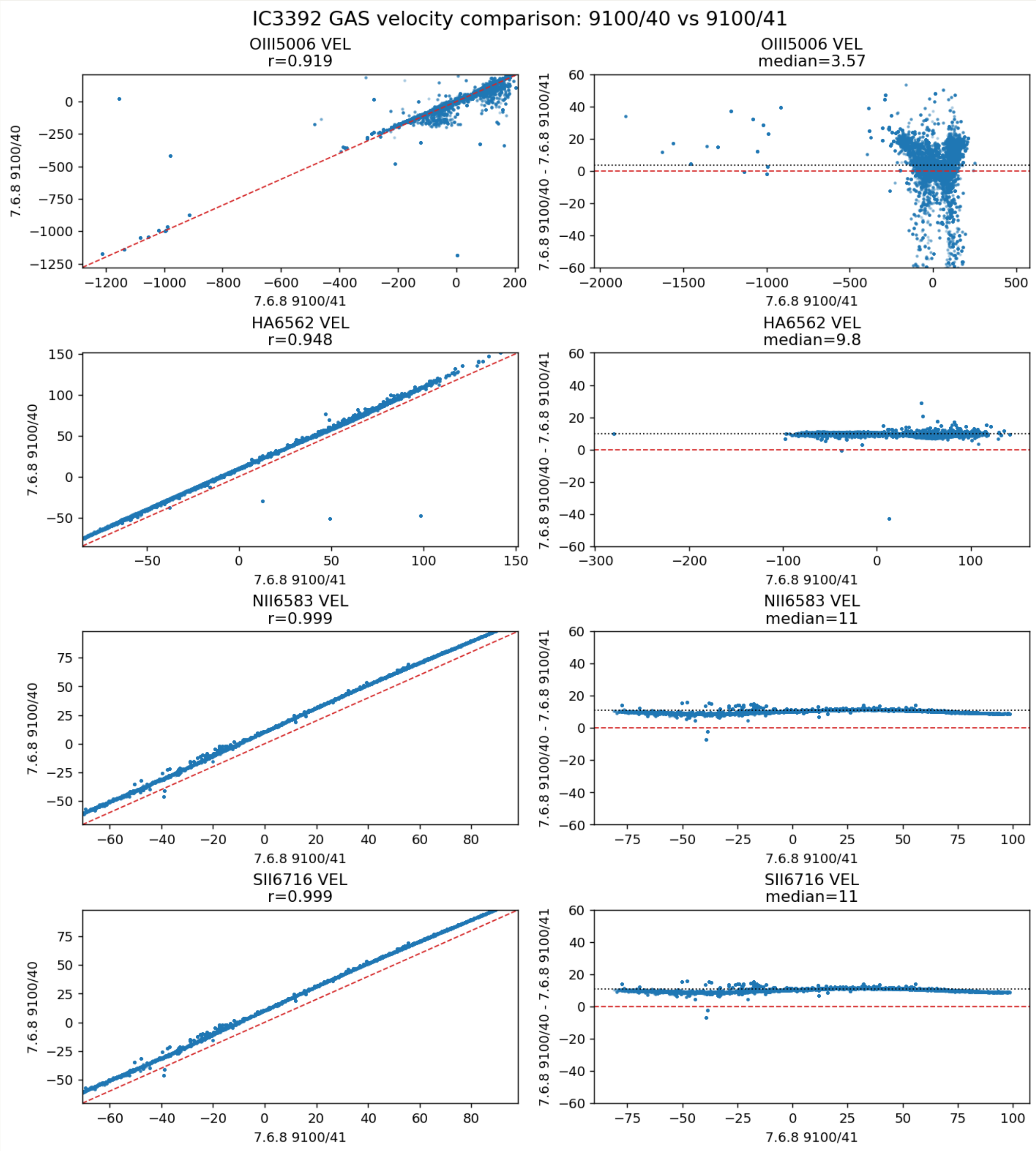
First, recall that `v768_8900_41` shows nice agreements with `old_v721_7000`: all gas velocity offsets are $-3 \sim -4$ km/s, which are similar to stellar velocity offset. But keep an eye on [O III] line (also for the following comparison plots), it becomes much more messy in pairwise scatter plot and the deviation is slightly higher. This is probably due to we tie [S III] line's kinematics with it.



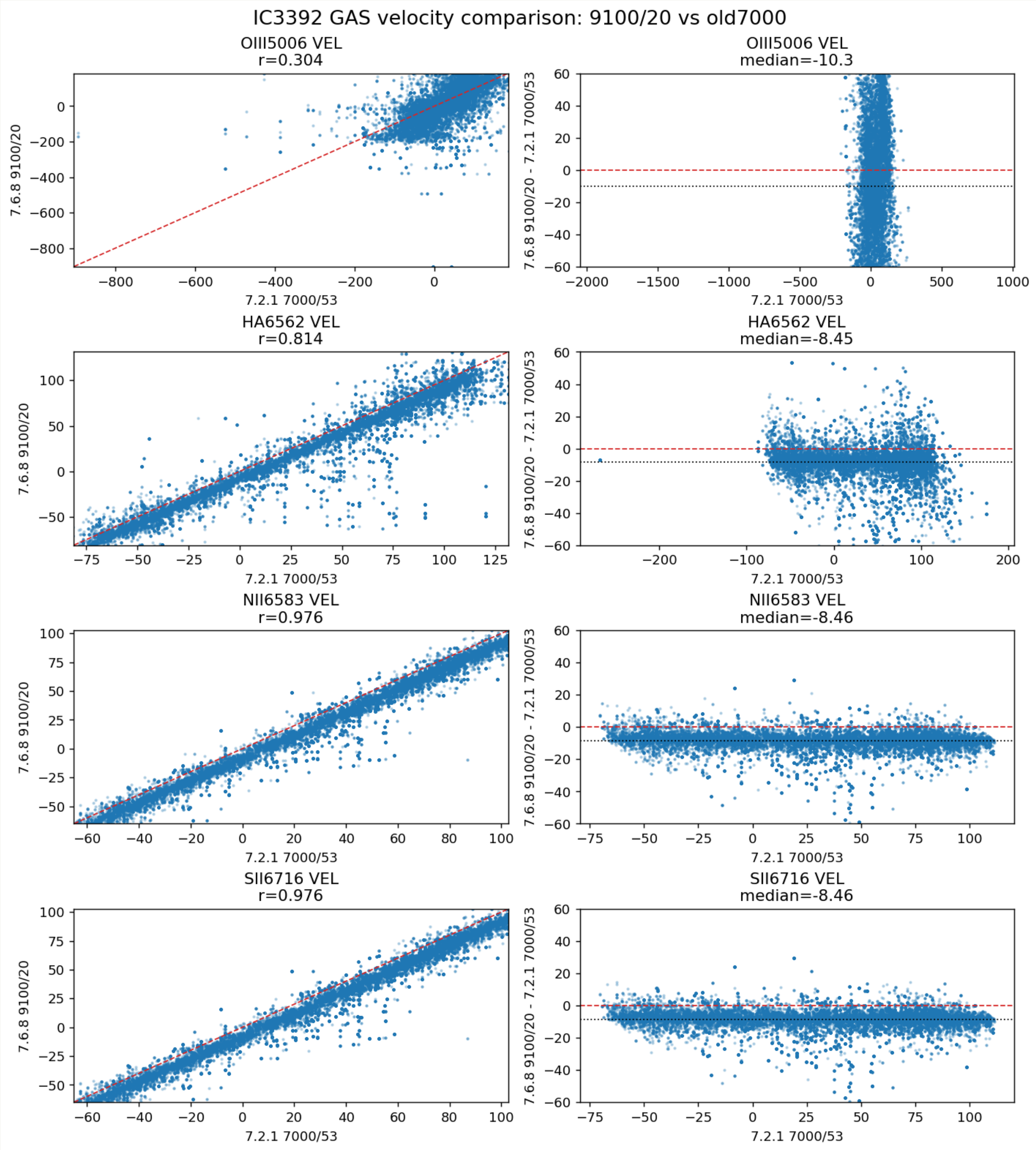
Then, if we take `v768_9100_40`, it also shows nice agreements with the old run.



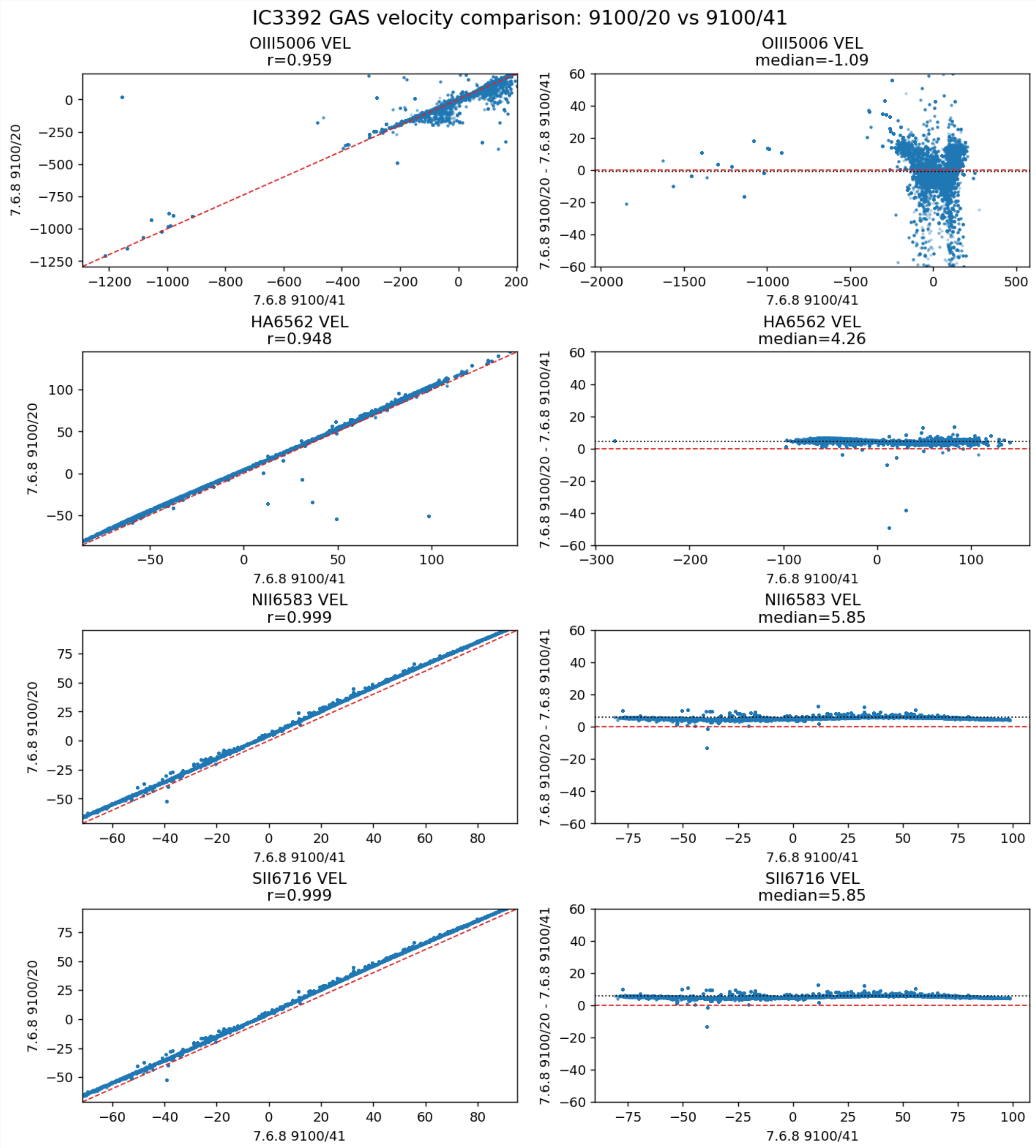
We also compare `v768_9100_40` with `v768_9100_41`. The [O III] line fitting is kind of self-consistent here, so maybe confirms that including [S III] fitting is risky.



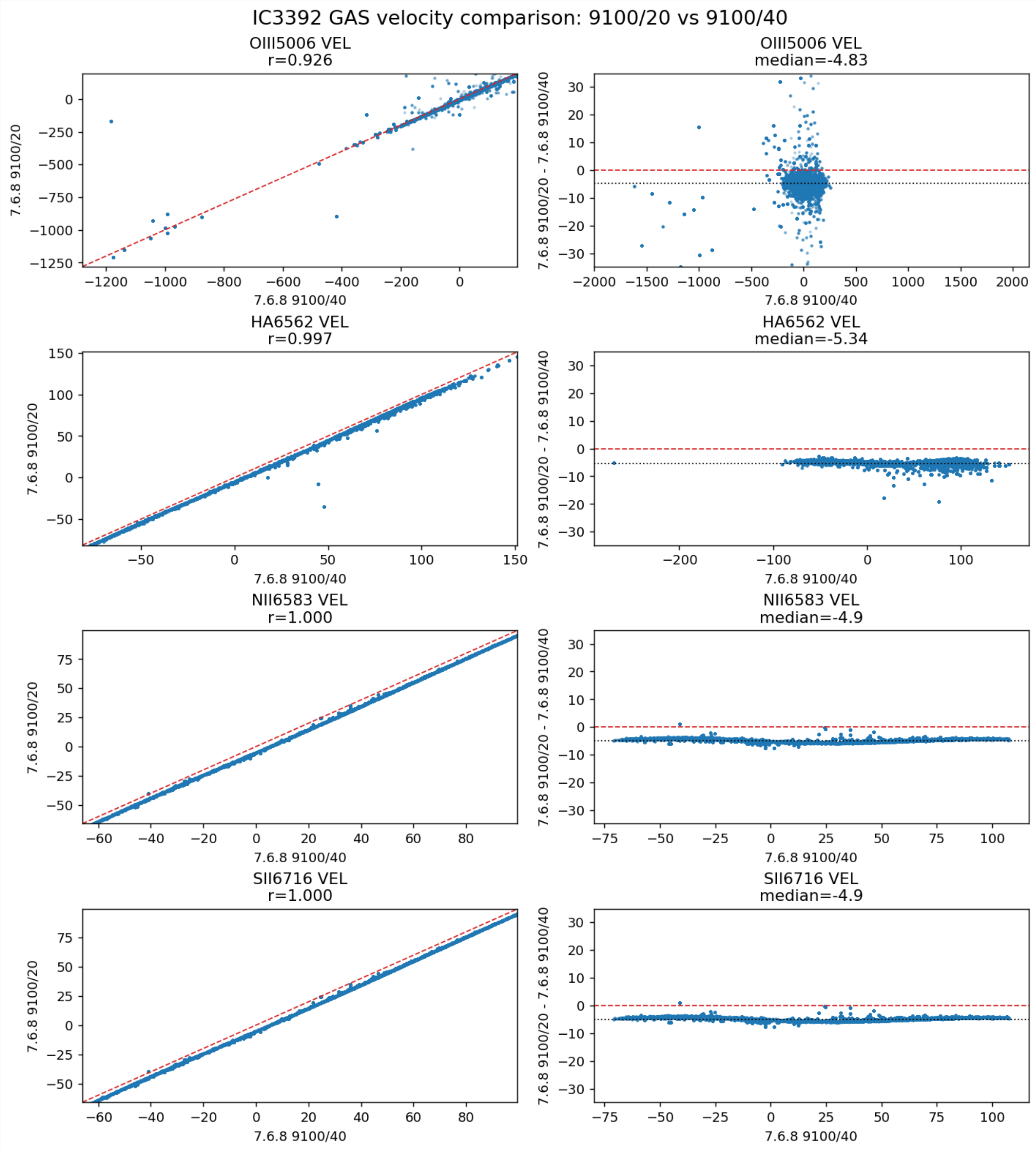
Finally, we show the oversampling `v768_9100_20` vs `old_v721_7000`. I am surprised to see that it improves relative to the problematic `9100/41`, **but** it does not converge onto the better `v768_9100_20` solution. So oversampling alone is not monotonically better and the actual log-rebin placement still matters.



Additionally, we have `v768_9100_20` vs `v768_9100_41` and `v768_9100_20` vs `v768_9100_40` in the end.



IC3392 GAS velocity comparison: 9100/20 vs 9100/41
OIII5006 VEL
r=0.959
OIII5006 VEL
median=-1.09
HA6562 VEL
r=0.948
HA6562 VEL
median=4.26
NII6583 VEL
r=0.999
NII6583 VEL
median=5.85
SII6716 VEL
r=0.999
SII6716 VEL
median=5.85
7.6.8 9100/20
7.6.8 9100/20 - 7.6.8 9100/41
7.6.8 9100/41


IC3392 GAS velocity comparison: 9100/20 vs 9100/40
OIII5006 VEL
r=0.926
OIII5006 VEL
median=-4.83
HA6562 VEL
r=0.997
HA6562 VEL
median=-5.34
NII6583 VEL
r=1.000
NII6583 VEL
median=-4.9
SII6716 VEL
r=1.000
SII6716 VEL
median=-4.9
7.6.8 9100/20
7.6.8 9100/40
7.6.8 9100/20 - 7.6.8 9100/40